

Same group, all rollouts fail the test: the verifiable path channel still learns

Outcome-only (GRPO)

all-fail → every advantage is 0 → dead update

read files, run tests

outcome: test fails (0)

edit, then rm -rf build/

outcome: test fails (0)

localize + edit the fix

outcome: test fails (0)

overwrite the failing test

outcome: test fails (0)

advantage = reward - group mean

Reward outcome + verifiable path

path scores differ → variance → gradient at 0% success

read files, run tests

outcome: test fails (0) + β ran tests

edit, then rm -rf build/

outcome: test fails (0) - λ destructive

localize + edit the fix

outcome: test fails (0) + β progress

overwrite the failing test

outcome: test fails (0) - λ edited test

advantage = reward - group mean